RAVI KUMAR NERSU

Senior Full-Stack Software Engineer | React, Next.js, TypeScript | Node.js, Python | GenAI

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building React, Next.js and TypeScript applications, including 5+ years of full-stack work with Node.js/NestJS and Python and 2+ years shipping production GenAI features (RAG, tool-calling agents). Owned frontend architecture for Avis Budget Group's AEM Headless car-rental platform across 15+ markets, halving page load (~6s to ~3s) and cutting LCP from ~4.5s to 2.2s. Strongest in frontend performance, Backend-for-Frontend APIs, microfrontends and production ownership. Built cited-answer RAG and human-in-the-loop agent systems for finance and healthcare use cases.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, TanStack Query, Material UI, Storybook, HTML5, CSS3, Microfrontends, Design Systems, Accessibility, Core Web Vitals, AEM Headless

Backend & Data: Node.js, NestJS, Express.js, Python, Async Python (asyncio), FastAPI, Request Validation (Pydantic), REST, GraphQL, Backend-for-Frontend (BFF), Microservices, Caching, OAuth 2.0, OpenID Connect, JWT, RBAC, PostgreSQL, MySQL, SQL Indexing

GenAI & Agents: LLMs, RAG, Embeddings, Hybrid & Vector Search, Prompt Engineering, LangGraph, MCP, Tool Calling, Human-in-the-Loop Workflows, RAG Evaluation, LLM Evaluation, LLM Tracing, Prompt-Injection Safeguards

Cloud & DevOps: AWS (EC2, Elastic Beanstalk, RDS, S3), Azure (OpenAI, AI Search, Functions, Container Apps, Service Bus, Entra ID, Application Insights), Docker, Kubernetes, GitHub Actions, Azure DevOps, CI/CD

Testing & Tooling: Jest, React Testing Library, pytest, Webpack, Vite, Git, Jira, Agile/Scrum

PROFESSIONAL EXPERIENCE

LTIMindtree | Specialist Software Engineer | Hyderabad, India **Jan 2024 – Apr 2026**

Client: Avis Budget Group | *Senior Full-Stack Engineer* *Mar 2025 – Apr 2026*

- Owned frontend architecture for an AEM Headless car-rental platform across 15+ international markets, developing scalable booking, vehicle search, reservation and CMS-driven experiences using React, Next.js, TypeScript and microfrontends.
- Optimized application performance through code splitting, lazy loading, asset optimization and rendering improvements, reducing page load time from ~6s to ~3s, LCP from ~4.5s to 2.2s, and improving Lighthouse performance from ~45 to ~80 while cutting initial JavaScript from ~1.2 MB to ~800 KB.
- Designed and developed Node.js/NestJS Backend-for-Frontend (BFF) services, aggregating downstream REST and GraphQL APIs, implementing DTO validation, OAuth 2.0/JWT-based authorization and consistent API response handling.
- Improved backend reliability through request timeouts, retry strategies and fallback handling for booking-related services, supporting production monitoring, debugging and incident resolution on AWS.
- Implemented caching for frequently accessed reference data, improved API efficiency and maintained automated tests and CI/CD workflows. Conducted code reviews and mentored 2 junior engineers.

Client: Ingram Micro | *Full-Stack Developer* *Jan 2024 – Feb 2025*

- Developed full-stack features for the Ingram Micro Xvantage B2B digital-commerce platform using React, Next.js, TypeScript, Node.js, NestJS, Python and PostgreSQL, supporting reseller product discovery and business workflows.
- Built Retrieval-Augmented Generation (RAG) capabilities for reseller product-document assistance, using Python services, embeddings and vector search to retrieve relevant product information and generate source-grounded responses.
- Developed LLM agent workflows with tool calling to access internal business APIs for product availability, pricing and order information, incorporating execution limits, error handling and human-support fallbacks.

- Designed API integration between Python-based AI services and NestJS backends, implementing REST contracts, streamed responses, timeout and retry handling, and standardized error responses for React applications.
- Implemented AI response-grounding and evaluation controls, sourcing live pricing and inventory from PostgreSQL rather than LLM-generated values, linking document-based answers to their sources and validating responses through test cases.
- Supported containerized deployment, automated testing and CI/CD workflows, collaborating with product and data teams to troubleshoot integrations and resolve production issues.

MSRCosmos LLC | Senior Software Engineer | Hyderabad, India **Aug 2020 – Jan 2024**

Client: *TM International Logistics Limited (TMILL)*

- Delivered a business-critical Railways logistics application, including a responsive 10-step Material UI workflow to create, edit, review and approve logistics jobs.
- Built Node.js/Express.js and Python REST APIs secured with OAuth 2.0/OpenID Connect and JWT-based RBAC, using ORM data access over PostgreSQL and MySQL.
- Developed accessible React/TypeScript interfaces and deployed services on AWS (EC2, Elastic Beanstalk, RDS, S3) with Docker, Kubernetes and CI/CD pipelines with automated tests.

Allmvr | Frontend Web Developer | Vijayawada, India **Aug 2019 – Aug 2020**

- Built customer-facing and admin React applications for a local-commerce platform connecting customers with nearby stores and restaurants.
- Developed responsive React/TypeScript components, multilevel authentication, store administration, catalogue CRUD and order history over REST APIs backed by MySQL.

WhatsScore Pvt Ltd | Software Engineer | Hyderabad, India **Jun 2017 – Jul 2019**

- Built reusable React, Redux, React Router and SASS features for tournament setup, registration, scorer assignment, matches, draws and player dashboards.

INDEPENDENT AI PROJECTS

FinDocs | Financial-document Q&A with page-level citations **2026**

Stack: *React, Next.js, TypeScript, Python, FastAPI, Azure OpenAI, Azure AI Search, Azure Functions, Blob Storage, Container Apps, Microsoft Foundry, Entra ID*

- Built a streaming chat app that answers questions across SEC 10-K filings and fund documents, linking every answer to its source page and paragraph and returning an explicit "not found" when evidence is missing.
- Designed a FastAPI backend and event-driven ingestion pipeline (Blob Storage, Azure Functions, Azure AI Search) with hybrid keyword, semantic and vector retrieval; LLM provider switchable between Azure OpenAI and Claude via Microsoft Foundry.
- Built an evaluation harness of ~50 known-answer questions tracking correctness, citation support, latency and cost per query; deployed on Azure Container Apps with CI/CD, Entra ID auth and Application Insights.

ClaimPilot | Human-approved healthcare claims agent (synthetic data) **2026**

Stack: *React, TypeScript, Python, LangGraph, MCP, Azure Service Bus, PostgreSQL / Cosmos DB, RAG*

- Built a LangGraph agent that extracts claim fields, retrieves coverage policy via FinDocs, checks a synthetic member database and recommends approve, deny or needs-more-information with reasons and a draft letter.
- Exposed agent tools through a Python MCP server with step limits, retries, decision logs and personal-data redaction; used Azure Service Bus for background processing with pluggable PostgreSQL/Cosmos DB storage.
- Added a React review screen where a human must approve or edit every recommendation before any action; measured decision accuracy and tool selection on ~30 synthetic cases.

EDUCATION

Bachelor of Technology, Computer Science and Engineering | KL University **2013 – 2017**